



SECP2523: DATABASE SEM I 2024/2025

DATABASE DESIGN: FOR KOPERASI KAKITANGAN KADA (KKK) ONLINE SYSTEM USER AUTHENTICATION AND REGISTRATION MODULE

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SQL Statement

Data Definition Language (DDL)

tb_member

```
1  -- Table structure for table `tb_member`
2  --
3
4
5  CREATE TABLE `tb_member` (
6
7      `m_memberApplicationID` int(11) NOT NULL,
8      `m_name` varchar(50) NOT NULL,
9      `m_ic` varchar(14) NOT NULL,
10     `m_email` varchar(255) NOT NULL,
11     `m_gender` int(11) NOT NULL,
12     `m_religion` int(11) NOT NULL,
13     `m_race` int(11) NOT NULL,
14     `m_maritalStatus` int(11) NOT NULL,
15     `m_homeAddress` varchar(50) NOT NULL,
16     `m_homePostcode` int(11) NOT NULL,
17     `m_homeCity` varchar(50) NOT NULL,
18     `m_homeState` int(11) NOT NULL,
19     `m_memberNo` int(11) DEFAULT NULL,
20     `m_pfNo` int(11) DEFAULT NULL,
21     `m_position` varchar(50) NOT NULL,
22     `m_positionGrade` varchar(11) NOT NULL,
23     `m_officeAddress` varchar(50) NOT NULL,
24     `m_officePostcode` int(11) NOT NULL,
25     `m_officeCity` varchar(50) NOT NULL,
26     `m_officeState` int(11) NOT NULL,
27     `m_phoneNumber` varchar(13) NOT NULL,
28     `m_homeNumber` varchar(13) DEFAULT NULL,
29     `m_monthlySalary` int(11) NOT NULL,
30     `m_feeMasuk` double NOT NULL,
31     `m_modalSyer` double NOT NULL,
32     `m_modalYuran` double NOT NULL,
33     `m_deposit` double NOT NULL,
34     `m_alAbrar` double NOT NULL,
35     `m_simpananTetap` double NOT NULL COMMENT 'Sumbangan Tabung
36     Tabung Kebajikan ',
37     `m_feeLain` double NOT NULL,
38
39
40
41
42
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45
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47
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49
50
```

```

51     `m_status` int(11) NOT NULL,
52     `m_applicationDate` datetime DEFAULT NULL,
53     `m_approvalDate` timestamp NULL DEFAULT NULL,
54     `m_adminID` int(11) DEFAULT NULL COMMENT 'Admin who approve
55 the application'
56 )
57 -- Indexes for table `tb_member`
58 --
59 ALTER TABLE `tb_member`
60     ADD PRIMARY KEY (`m_memberApplicationID`),
61     ADD UNIQUE KEY `m_memberApplicationID`
62     (`m_memberApplicationID`),
63     ADD UNIQUE KEY `m_memberNo` (`m_memberNo`),
64     ADD UNIQUE KEY `m_memberNo_2` (`m_memberNo`),
65     ADD KEY `m_gender`
66     (`m_gender`,`m_religion`,`m_race`,`m_maritalStatus`,`m_status`),
67     ADD KEY `m_religion` (`m_religion`),
68     ADD KEY `m_race` (`m_race`),
69     ADD KEY `m_maritalStatus` (`m_maritalStatus`),
70     ADD KEY `m_status` (`m_status`),
71     ADD KEY `m_adminID` (`m_adminID`),
72     ADD KEY `m_homeState` (`m_homeState`,`m_officeState`),
73     ADD KEY `tb_member_ibfk_7` (`m_officeState`);
74 --
75 -- AUTO_INCREMENT for table `tb_member`
76 --
77 ALTER TABLE `tb_member`
78     MODIFY `m_memberApplicationID` int(11) NOT NULL
79     AUTO_INCREMENT;
80 --
81 -- Constraints for table `tb_member`
82 --
83 ALTER TABLE `tb_member`
84     ADD CONSTRAINT `tb_member_ibfk_1` FOREIGN KEY (`m_gender`)
85     REFERENCES `tb_ugender` (`ug_gid`),
86     ADD CONSTRAINT `tb_member_ibfk_2` FOREIGN KEY (`m_religion`)
87     REFERENCES `tb_ureligion` (`ua_rid`),
88     ADD CONSTRAINT `tb_member_ibfk_3` FOREIGN KEY (`m_race`)
89     REFERENCES `tb_urace` (`ur_rid`),
90
91
92
93
94
95
96
97
98
99
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101
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103
104
105
106

```

```

107     ADD CONSTRAINT `tb_member_ibfk_4` FOREIGN KEY
108     (`m_maritalStatus`) REFERENCES `tb_umaritalstatus` (`um_mid`),
109     ADD CONSTRAINT `tb_member_ibfk_5` FOREIGN KEY (`m_status`)
110     REFERENCES `tb_status` (`s_sid`),
111     ADD CONSTRAINT `tb_member_ibfk_6` FOREIGN KEY (`m_homeState`)
112     REFERENCES `tb_homestate` (`st_id`),
113     ADD CONSTRAINT `tb_member_ibfk_7` FOREIGN KEY
114     (`m_officeState`) REFERENCES `tb_officestate` (`st_id`);
115
116
117
118
119

```

tb_heir

```

1  CREATE TABLE `tb_heir` (
2      `h_heirID` int(11) NOT NULL,
3      `h_memberApplicationID` int(11) NOT NULL,
4      `h_name` varchar(50) NOT NULL,
5      `h_relationWithMember` int(10) NOT NULL,
6      `h_ic` varchar(14) NOT NULL
7  )
8
9  -- Indexes for table `tb_heir`
10 --
11
12 ALTER TABLE `tb_heir`
13     ADD PRIMARY KEY (`h_heirID`),
14     ADD KEY `h_memberApplicationID` (`h_memberApplicationID`),
15     ADD KEY `h_relationWithMember` (`h_relationWithMember`);
16
17 -- AUTO_INCREMENT for table `tb_heir`
18 --
19 ALTER TABLE `tb_heir`
20     MODIFY `h_heirID` int(11) NOT NULL AUTO_INCREMENT,
21     AUTO_INCREMENT=10;
22
23 -- Constraints for table `tb_heir`
24 --
25 ALTER TABLE `tb_heir`
26     ADD CONSTRAINT `tb_heir_ibfk_1` FOREIGN KEY
27     (`h_memberApplicationID`) REFERENCES `tb_member`
28     (`m_memberApplicationID`),
29
30
31
32
33
34
35
36
37
38

```

```
39 ADD CONSTRAINT `tb_heir_ibfk_2` FOREIGN KEY
40 (`h_relationWithMember`) REFERENCES `tb_hrelation` (`hr_rid`);
41
```

tb_homestate

```
1 CREATE TABLE `tb_homestate` (
2   `st_id` int(11) NOT NULL,
3   `st_desc` varchar(15) NOT NULL
4 )
5
6
7
8 -- Indexes for table `tb_homestate`
9 --
10
11 ALTER TABLE `tb_homestate`
12   ADD PRIMARY KEY (`st_id`);
13
```

tb_hrelation

```
1 CREATE TABLE `tb_hrelation` (
2   `hr_rid` int(11) NOT NULL,
3   `hr_desc` varchar(20) NOT NULL
4 )
5
6
7
8 -- Indexes for table `tb_hrelation`
9 --
10
11 ALTER TABLE `tb_hrelation`
12   ADD PRIMARY KEY (`hr_rid`);
13
```

tb_officestate

```
1 CREATE TABLE `tb_officestate` (
2   `st_id` int(11) NOT NULL,
3   `st_desc` varchar(15) NOT NULL
4 )
5
6
7
8 -- Indexes for table `tb_officestate`
9 --
10
11 ALTER TABLE `tb_officestate`
12   ADD PRIMARY KEY (`st_id`);
```

tb_policies

```
1 CREATE TABLE `tb_policies` (  
2     `p_policyID` int(10) NOT NULL,  
3     `p_memberRegFee` double NOT NULL COMMENT 'Fee Masuk',  
4     `p_minShareCapital` double NOT NULL COMMENT 'Modah Syer  
5 Minimum',  
6     `p_minFeeCapital` double NOT NULL COMMENT 'Modal Yuran  
7 Minimum',  
8     `p_minFixedSaving` double NOT NULL COMMENT 'Wang deposit  
9 Anggota Minimum',  
10    `p_minMemberFund` double NOT NULL COMMENT 'Sumbangan Tabung  
11 Kebajikan Minimum',  
12    `p_minMemberSaving` double NOT NULL COMMENT 'Simpanan Tetap  
13 Minimum',  
14    `p_minOtherFees` double NOT NULL COMMENT 'Lain-lain',  
15    `p_minShareCapitalForLoan` double NOT NULL COMMENT 'Modah Syer  
16 Minimum',  
17    `p_profitRate` double NOT NULL COMMENT 'Kadar Keuntungan',  
18    `p_maxInstallmentPeriod` int(11) NOT NULL COMMENT 'Tempoh  
19 Ansuran Maksima',  
20    `p_maxFinancingAmt` double NOT NULL COMMENT 'Jumlah Pembiayaan  
21 Maksima',  
22    `p_salaryDeductionForSaving` double NOT NULL COMMENT 'Potongan  
23 Gaji Bulanan untuk Simpanan Tetap',  
24    `p_salaryDeductionForMemberFund` double NOT NULL COMMENT  
25 'Potongan Gaji Bulanan untuk Tabung Kebajikan',  
26    `p_dateUpdated` timestamp NOT NULL DEFAULT current_timestamp()  
27 ON UPDATE current_timestamp(),  
28    `p_adminID` int(11) NOT NULL  
29 )  
30  
31 -- Indexes for table `tb_policies`  
32 --  
33 ALTER TABLE `tb_policies`  
34     ADD PRIMARY KEY (`p_policyID`),  
35     ADD KEY `p_adminID` (`p_adminID`);  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52
```

```

53 -- AUTO_INCREMENT for table `tb_policies`
54 --
55
56 ALTER TABLE `tb_policies`
57     MODIFY `p_policyID` int(10) NOT NULL AUTO_INCREMENT,
58     AUTO_INCREMENT=2;
59
60
61 -- Constraints for table `tb_policies`
62 --
63
64 ALTER TABLE `tb_policies`
65     ADD CONSTRAINT `tb_policies_ibfk_1` FOREIGN KEY (`p_adminID`)
66     REFERENCES `tb_admin` (`a_adminID`);
67

```

tb_ugender

```

1 CREATE TABLE `tb_ugender` (
2     `ug_gid` int(11) NOT NULL,
3     `ug_desc` varchar(10) NOT NULL
4 )
5
6
7
8 -- Indexes for table `tb_ugender`
9 --
10
11 ALTER TABLE `tb_ugender`
12     ADD PRIMARY KEY (`ug_gid`);
13

```

tb_urace

```

1 CREATE TABLE `tb_urace` (
2     `ur_rid` int(11) NOT NULL,
3     `ur_desc` varchar(10) NOT NULL
4 )
5
6
7
8 -- Indexes for table `tb_urace`
9 --
10
11 ALTER TABLE `tb_urace`
12     ADD PRIMARY KEY (`ur_rid`);
13

```

tb_ureligion

```

1 CREATE TABLE `tb_ureligion` (
2   `ua_rid` int(11) NOT NULL,
3   `ua_desc` varchar(10) NOT NULL
4 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
5 COLLATE=utf8mb4_general_ci;
6
7 -- Indexes for table `tb_ureligion`
8 --
9
10 ALTER TABLE `tb_ureligion`
11   ADD PRIMARY KEY (`ua_rid`);
12
13
14
15

```

tb_user

```

1 CREATE TABLE `tb_user` (
2   `u_id` int(11) NOT NULL,
3   `reset_token` varchar(255) NOT NULL,
4   `token_expiry` timestamp NOT NULL DEFAULT current_timestamp()
5   ON UPDATE current_timestamp(),
6   `u_pwd` varchar(255) DEFAULT NULL,
7   `u_type` int(11) NOT NULL
8 )
9
10 -- Indexes for table `tb_user`
11 --
12
13 ALTER TABLE `tb_user`
14   ADD PRIMARY KEY (`u_id`),
15   ADD KEY `u_type` (`u_type`);
16
17
18
19
20
21

```

tb_ctype

```

1 CREATE TABLE `tb_ctype` (
2   `ct_tid` int(11) NOT NULL,
3   `ct_desc` varchar(10) NOT NULL
4 )
5
6 -- Indexes for table `tb_ctype`
7
8
9

```



```

10 --
11 ALTER TABLE `tb_utype`
12     ADD PRIMARY KEY (`ut_tid`);
13
14

```

Data Manipulation Language (DML)

Use Case: Log In

```

1  //Retrieves data from tb_user for user availability by user id
2
3  $sql = "SELECT * FROM tb_user WHERE u_id = ?";
4  $stmt = mysqli_prepare($con, $sql);
5  mysqli_stmt_bind_param($stmt, 'i', $fname);
6  mysqli_stmt_execute($stmt);
7  $result = mysqli_stmt_get_result($stmt);
8  $user = mysqli_fetch_assoc($result);
9  $matches = false;
10
11  if($user){
12      $matches = password_verify($fpwd, $user['u_pwd']);
13  }
14  //Retrieves data from tb_member for matching memberId and email
15
16  $sql = "SELECT m_memberNo, m_email
17          FROM tb_member
18          WHERE m_memberNo = ? AND m_email = ?";
19  $stmt = mysqli_prepare($con, $sql);
20  mysqli_stmt_bind_param($stmt, 'is', $userID, $email);
21  mysqli_stmt_execute($stmt);
22  $result = mysqli_stmt_get_result($stmt);
23
24  if (mysqli_num_rows($result) > 0) {
25      //generate a unique token that last 10 min
26      $token = bin2hex(random_bytes(16));
27      $expiryTime = date("Y-m-d H:i:s", strtotime("+10 minutes",
28      time()));
29
30
31  //Set reset_token and expiry time in tb_user by userID
32
33      $sql = "UPDATE tb_user SET reset_token = ?, token_expiry = ?
34      WHERE u_id = ?";
35      $stmt = mysqli_prepare($con, $sql);
36      mysqli_stmt_bind_param($stmt, 'ssi', $token, $expiryTime,
37      $userID);
38      mysqli_stmt_execute($stmt);
39
40  //check for matching token and expiry status in tb_user by reset token
41

```

```

42     $sql = "SELECT * FROM tb_user WHERE reset_token = ?";
43     $stmt = mysqli_prepare($con, $sql);
44     mysqli_stmt_bind_param($stmt, 's', $token);
45     mysqli_stmt_execute($stmt);
46     $result = mysqli_stmt_get_result($stmt);
47     $user = mysqli_fetch_assoc($result);
48
49 //reset password by reset token
50     $sql = "UPDATE tb_user SET u_pwd = ?, reset_token = NULL,
51 token_expiry = NULL WHERE reset_token = ?";
52     $stmt = mysqli_prepare($con, $sql);
53     mysqli_stmt_bind_param($stmt, 'ss', $hashedPassword, $token);
54     mysqli_stmt_execute($stmt);
55
56
57
58

```

Use Case: Apply for Membership

```

1 //finds for similar ic and email in tb_member to check for
2 duplicated entries
3 $fuic = $cleanPost['fuic'];
4 $sql = "SELECT * FROM tb_member WHERE m_ic = ?";
5 $binds = [$fuic];
6 $result = query($sql, $binds);
7
8 $row = mysqli_fetch_assoc($result);
9
10 if ($row) {
11 //display error message
12 }
13
14 $femail = $cleanPost['femail'];
15 $sql_email = "SELECT * FROM tb_member WHERE m_email = ?";
16 $binds_email = [$femail];
17 $result_email = query($sql_email, $binds_email);
18
19
20 if ($row_email = mysqli_fetch_assoc($result_email)) {
21     display error message
22 }
23
24 //retrieves a single latest minimum value of fees from tb_policies
25
26 $minValues = [];
27 $query = "
28     SELECT
29         p_memberRegFee AS ffee,
30         p_minShareCapital AS fmodal,
31         p_minFeeCapital AS fyuran,
32         p_minFixedSaving AS fanggota,
33         p_minMemberFund AS fabrar,
34         p_minMemberSaving AS ftetap,
35         p_minOtherFees AS fother

```



```

36     FROM tb_policies
37     ORDER BY p_policyID DESC
38     LIMIT 1";
39
40 $result = mysqli_query($con, $query);
41 if ($result && $row = mysqli_fetch_assoc($result)) {
42     $minValues = $row;
43 }
44
45 //registration (insert into database)
46 include('../db_connect.php');
47     $fdate = date('Y-m-d H:i:s');
48
49     // Start transaction
50     $con->begin_transaction();
51
52     // Insert into tb_member
53     $stmt = $con->prepare("
54         INSERT INTO tb_member (
55             m_name, m_ic, m_email, m_gender, m_religion, m_race,
56             m_maritalStatus, m_homeAddress, m_homeCity, m_homeState,
57 m_homePostcode,
58             m_position, m_positionGrade, m_pfNo, m_officeAddress,
59 m_officeCity, m_officeState, m_officePostcode,
60             m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk,
61 m_modalSyer, m_modalYuran, m_deposit,
62             m_alAbrar, m_simpananTetap, m_feeLain, m_applicationDate,
63 m_status
64         ) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?,
65 ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
66     ");
67
68     $stmt->bind_param(
69         "sssssssssssssssssssssssssssssi",
70         $funame,
71         $fuic,
72         $femail,
73         $fgender,
74         $freligion,
75         $frace,
76         $fmariage,
77         $fhomeaddress,
78         $fcity,
79         $fstate,
80         $fhomezip,
81         $fposition,
82         $fgrade,
83         $fpfno,
84         $fofficeaddress,
85         $fofficecity,
86         $fofficestate,
87         $fofficezip,
88         $ftelno,
89         $fhomeno,
90         $fsalary,
91         $ffee,

```

```

92         $fmodal,
93         $fyuran,
94         $fanggota,
95         $fabrar,
96         $ftetap,
97         $fother,
98         $fdate,
99         $fstatus
100     );
101
102     if ($stmt->execute()) {
103         // Get the last inserted member ID
104         $member_id = $con->insert_id;
105
106         // Prepare heir insert statement
107         $stmt_heir = $con->prepare("
108             INSERT INTO tb_heir (h_memberApplicationID, h_name, h_ic,
109 h_relationWithMember)
110             VALUES (?, ?, ?, ?)
111         ");
112
113         if ($stmt_heir) {
114             // Insert heirs
115             foreach ($heirs as $heir) {
116                 $stmt_heir->bind_param("isss", $member_id,
117 $heir['name'], $heir['ic'], $heir['relation']);
118                 if (!$stmt_heir->execute()) {
119                     // Rollback on error
120                     $con->rollback();
121                     echo "<script>alert('Error inserting heir data: "
122 . $stmt_heir->error . "')</script>";
123                     exit();
124                 }
125             }
126
127             // Commit the transaction
128             $con->commit();
129
130
131 //display information
132 if (isset($_GET['member_id']) && !empty($_GET['member_id'])) {
132     $member_id = intval($_GET['member_id']); // Sanitize input
133
134     // Retrieve member details from tb_member by memberID for display
135     purpose
136     $stmt = $con->prepare("SELECT * FROM tb_member
137     LEFT JOIN tb_ugender ON tb_member.m_gender = tb_ugender.ug_gid
138     LEFT JOIN tb_ureligion ON tb_member.m_religion =
139 tb_ureligion.ua_rid
140     LEFT JOIN tb_urace ON tb_member.m_race = tb_urace.ur_rid
141     LEFT JOIN tb_homestate ON tb_member.m_homeState =
142 tb_homestate.st_id
143     LEFT JOIN tb_officestate ON tb_member.m_officeState =
144 tb_officestate.st_id
145     WHERE m_memberApplicationID = ?");
146     $stmt->bind_param("i", $member_id);

```